2. Data and AI in Private Markets

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Data and AI in Finance

- Data ⇒ Prediction ⇒ Decision
- Credit markets

  Mortgages, consumer loans, business loans

  ⇒ Predict default

- Private equity

  Private equity, venture capital, project finance

  ⇒ Predict cash flow

- Insurance markets

  Health, property and casualty

  ⇒ Predict losses and damages

- Public markets (class 5)

  Asset management, market making

  ⇒ Predict future price

Road Map

Mini-Case 1: Credit Scoring

VC Game

Mini-Case 2: Zillow Offers

Statistical Underpinnings

Goodhart's Law

Discrimination and Redistributive Effects

Hirshleifer Effect

Credit Scoring with Digital Footprints: An Early Adopter

- E-commerce company in Germany, 2015[1]
- The company conducted A/B testing
  - Randomized customers viewing an item on the website
  - Treatment group: Offered the option to pay within 15 days of receiving the purchased item
  - Control group: Option not offered
  - Both groups had the same item price
- Impact
  - Control group: 45% probability of purchasing the item
  - Treated group: 85% probability of purchasing the item
- What should the company's management do?

[1] Berg, Burg, Gombovic, Puri, 2019, "On the Rise of FinTechs: Credit Scoring Using Digital Footprints" *Review of Financial Studies*

Credit Scoring with Digital Footprints

- Trade-off in offering payment facility: increased sales vs. default risk

$\Rightarrow$ Assess default risk and offer payment facility if default probability is low

- Phase 1: Purchase credit scores from a credit bureau
  - Scores based on credit history and sociodemographic data
  - Default rate: 3%
- Phase 2: Utilize proprietary customer data
  - Digital footprints: OS, email, login information, etc.
  - Does this data enhance default prediction?

Scoring Model

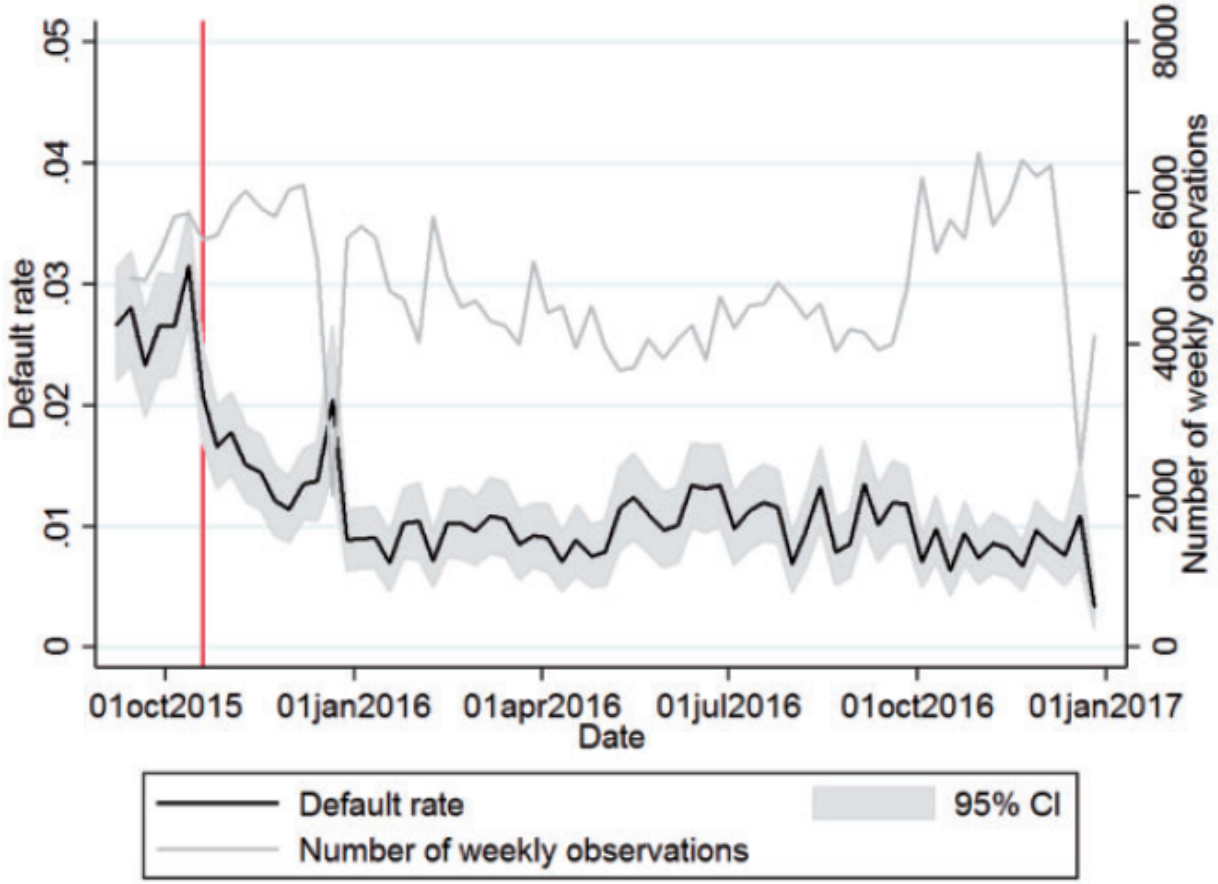
- Logistic regression to predict default
- Dependent variable: =1 if default
- Predictive variables
  - (1) Credit bureau score
  - (2) Digital footprints
  - (3) Credit bureau score + digital footprints

Default regressions (scorable customers)

	(1) Credit bureau bureau score		(2) Digital footprint		(3) Credit bureau score & digital footprint	
Variables	Coef.	z-stat	Coef.	z-stat	Coef.	z-stat
Credit bureau score	−0.17***	(−7.89)			−0.15***	(−6.67)
Device type & operating system[a]						
Desktop/Windows			Baseline		Baseline	
Desktop/Macintosh			−0.07	(−0.53)	−0.13	(−1.03)
Tablet/Android			0.29***	(3.19)	0.29***	(3.06)
Tablet/iOS			0.08	(1.05)	0.08	(0.97)
Mobile/Android			1.05***	(17.25)	0.95***	(15.34)
Mobile/iOS			0.72***	(9.07)	0.57***	(6.73)
E-mail Host [a]						
Gmx (partly paid)			Baseline		Baseline	
Web (partly paid)			0.00	(0.00)	−0.02	(−0.22)
T-Online (affluent customers)			−0.40***	(−3.90)	−0.35***	(−3.35)
Gmail (free)			0.34***	(3.81)	0.29***	(3.09)
Yahoo (free, older service)			0.75***	(9.19)	0.72***	(8.98)
Hotmail (free, older service)			0.35***	(3.70)	0.28***	(2.72)
Channel						
Paid			Baseline		Baseline	
Affiliate			−0.49***	(−5.35)	−0.54***	(−5.58)
Direct			−0.27***	(−4.25)	−0.28***	(−4.44)
Organic			−0.15*	(−1.79)	−0.15*	(−1.74)
Other			−0.47***	(−4.50)	−0.48***	(−4.36)
Checkout time						
Evening (6 p.m.-midnight)			Baseline		Baseline	
Morning (6 a.m.-noon)			0.28***	(4.50)	0.28***	(4.60)
Afternoon (noon-6 p.m.)			0.08	(1.42)	0.08	(1.47)
Night (midnight-6 a.m.)			0.79***	(7.73)	0.75***	(7.09)
Do-not-track setting			−0.02	(−0.25)	−0.07	(−0.91)
Name in e-mail			−0.28***	(−5.67)	−0.29***	(−5.70)
Number in e-mail			0.26***	(4.50)	0.23***	(3.91)
Is lowercase			0.76***	(13.10)	0.74***	(13.20)
E-mail error			1.66***	(20.00)	1.67***	(20.36)
Constant	12.42***	(5.76)	−4.92***	(−62.87)	9.97***	(4.48)
Control for *Age*, *Gender*, *Item category*, *Loan amount*, and month and region fixed effects	No		No		No	
Observations	254,819		254,819		254,819	
Pseudo R^2	**.0244**		**.0524**		**.0717**	
AUC	**0.683**		**0.696**		**0.736**	
(SE)	**(0.006)**		**(0.006)**		**(0.005)**	
Difference to AUC=50%	**0.183*** **		**0.196*** **		**0.236*** **	
Difference AUC to (1)			**0.013***		**0.053*** **	

Impact on Default Rate

- The new credit scoring model is put in production in October 2015
- Default rate divided by 3



Data and AI in Private Markets

- Using data and AI for prediction and decision-making can be very powerful
- But also important pitfalls to avoid
  - Adverse selection
  - Goodhart's Law
  - Discrimination and redistributive effects
  - Hirshleifer effect in insurance

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VC Game

- You manage a venture capital (VC) fund
- Potential deal: Acquire an ownership stake in a startup in exchange for cash payment P
- Discounted value of ownership stake is uncertain: V drawn from uniform distribution between 20 and 100
- You further analyze the company to refine your valuation $\Rightarrow$ your best estimate of V: check your private signal
- Your best estimate is unbiased: Generated as the true V plus Gaussian noise with mean 0 and s.d. 20

VC Game

- You compete with other VC funds (students) who also have their own estimate of V
- Each VC makes an offer P
- The highest offer wins the deal
- Your profit (or loss) is
  - $V - P$ if you win the deal at price P and the true value is V
  - zero if you don't win the deal
- Submit your bid at https://forms.gle/seYd1YQAkhLLe3WN6

Winner's Curse

- Estimates are unbiased, i.e., the average is $V \Rightarrow$ some are above, some are below $\Rightarrow$ the highest is above V
- If all bid their best estimate, the winning price is the highest estimate $\Rightarrow$ above $V \Rightarrow$ winner overpays
- This is the **winner's curse**

VC and AI

Q. What are other limitations of AI in the VC context?

VC and AI

Q. What are other limitations of AI in the VC context?

The Review of Financial Studies

Data-Driven Investors

Maxime Bonelli <== HEC PhD 2023
London Business School, United Kingdom

How does the increased use of data technologies, like machine learning, by financial intermediaries affect the allocation of capital towards innovation? I study this question in the context of startup financing by venture capitalists (VCs). While VCs adopting data technologies become better at screening startups similar to those in historical data, they tilt their investments towards this pool and become concurrently less likely to finance innovative startups that achieve rare major success. Plausibly exogenous variations in VCs' screening automation suggest that these effects are causal. These findings highlight how investors' adoption of data technologies can have real effects through innovation financing. (*JEL* G24, L26, O30)

Winner's Curse Everywhere

- **Lending:** Borrowers accept your loan offer when other lenders charge higher rates, indicating your credit risk estimate is too optimistic
- **Insurance:** Customers buy insurance from you when other insurers charge higher premiums, indicating your risk estimate is too optimistic
- **Trading:** See class 5
- Auctions, etc.
- The winner's curse is also called **adverse selection**: your offer is selected (accepted) only when it is too optimistic
- Multiple Nobel Prize-winning concept: 2001 (Akerlof, Stiglitz, Spence), 2020 (Milgrom, Wilson)

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Zillow Offers

- Zillow founded in 2006 in Seattle by former Expedia executives
- Initial core business: digital real-estate platform
  - Home search, listings, rentals, housing information
  - Revenue mainly from advertising and lead generation
  - Adjacent services: mortgages and related transaction services
- in 2018 Zillow launched **Zillow Offers** = "iBuyer" service
  - Homeowners request a direct offer from Zillow
  - Zillow buys the home, performs light repairs, and resells it

$\Rightarrow$ Zillow moved from platform intermediary to principal buyer and seller

Zillow Offers

- Operational steps

  (1) Seller requests an offer online

  (2) Zillow evaluates the home with data and AI and makes an offer

  (3) If accepted, Zillow buys the home and handles closing

  (4) Zillow performs light repairs

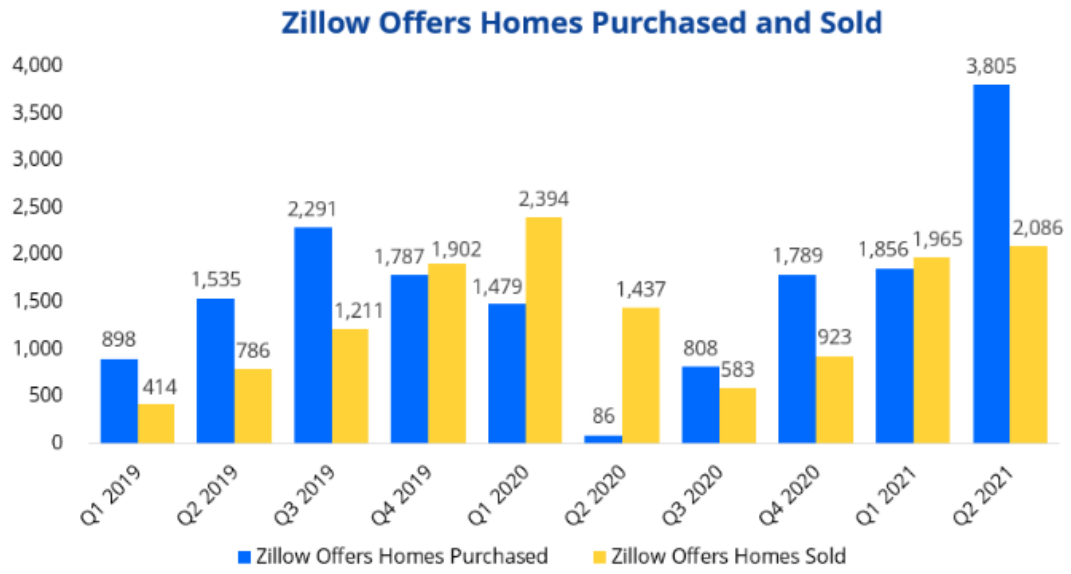
  (5) Zillow relists and resells the home

- Value proposition to sellers: speed, certainty, and convenience (avoid repeated showings)

Estimating Home Values

- Data used
  - Home characteristics: square footage, location, photos, etc.
  - On-market data: comparable homes' listing price, days on market
  - Off-market data: tax assessments, prior sales, public records
  - Market trends: local price dynamics and seasonal demand
- Model
  - Neural network model
  - From $1 million Kaggle competition on home valuation prediction
  - Input: see above
  - Output: estimated market value of the home
- Model performance: https://www.zillow.com/zestimate/

Financial Performance



Second-Quarter 2021 Zillow Offers Unit Economics

		Total		Average per Home	% of Revenue
Homes sold		2,086			
Zillow Offers revenue	$	772,030,000	$	370,101	100.0%
Operating costs:					
Home acquisition costs*		672,592,000		322,432	87.1%
Renovation costs*		20,862,000		10,001	2.7%
Holding costs*,**		4,527,000		2,170	0.6%
Selling costs		29,544,000		14,163	3.8%
Total operating costs		727,525,000		348,766	94.2%
Return on Homes Sold Before Interest Expense[3]		44,505,000		21,335	576 bps
Interest expense*,**		3,545,000		1,699	0.5%
Return on Homes Sold After Interest Expense[3]	$	40,960,000	$	19,636	531 bps

Zillow Offers EBITDA 2021Q2: negative $29 million (negative 4% of revenue)

Q. If gross margins on homes sold are positive, why is EBITDA negative?

Financial Performance

- EBITDA 2021Q3: negative $380 million (negative 32% of revenue)

- Shareholder letter 2021Q3:

  "We also recorded a $304 million write-down on inventory owned at the end of Q3 as a result of unintentionally purchasing homes at higher prices than our current estimates of future selling prices."

  "we have been able to convert only about 10% of the serious sellers who ask for a Zillow Offer, and have tended to disappoint the roughly 90% who didn't sell to us."

Q. What risks are revealed by the inventory write-down and the 10% seller conversion rate?

Adverse Selection

- Zillow's ideal business model
  - Zillow estimates the home's value at 100
  - Seller values speed and convenience at 4
  - Seller willing to sell at 96
  - Zillow offers 96, buys the home, and re-sells it at 100 (net of costs)
  - Margin = 4
- If the seller has information about the home's value that Zillow lacks
  - Zillow still estimates the home's value at 100
  - But the true value may be above or below 100
  - Sellers are more likely to accept when Zillow's offer is too high

  $\Rightarrow$ Zillow is more likely to buy overpriced homes

Adverse Selection

Zillow's estimate	100	100	100
Seller's value for speed and convenience	4	4	4
Zillow's offer	96	96	96
True value	100	90	110
Offer accepted?	Yes	Yes	No
Margin	4	Negative 6	0

Q. How to mitigate adverse selection?

(1) Bid more conservatively than the statistical estimate

- Create a margin of safety against overpaying
- But lower conversion rate

(2) Backtest after launch

- Compare forecast values with realized resale outcomes
- But reactive: losses may already be incurred

(3) Experiment with pricing on a small sample, then backtest

- Learn how offer levels affect acceptance, resale margins, and selection
- But slower scaling

(4) Focus on easy-to-value, liquid homes and markets

- Reduce hidden information and pricing error
- But weaker value proposition and narrower market

Road Map

Mini-Case 1: Credit Scoring

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Statistical Underpinnings

Goodhart's Law

Discrimination and Redistributive Effects

Hirshleifer Effect

Statistical Underpinnings of Adverse Selection

- Highly relevant for the group work
- You make loans to businesses
- Offer interest rate R, paid at issuance (to simplify formulas)
- Cash flow (per euro lent) if loan offer is accepted:

  At issuance: $-1 + R$

  At maturity: $\begin{cases} 1 \text{ if no default} \\ 0 \text{ if default} \end{cases} = 1 - D$ where $D \in \{0, 1\}$ is default indicator
- Assume zero discount rate. Profit $= (-1 + R) + (1 - D) = R - D$

Statistical Underpinnings of Adverse Selection

- Probability of Default: PD $= f(X) + U$
  - X: observed borrower characteristics ("features" in ML)
  - U: unobserved determinants of default risk that cannot be recovered from X. $\mathbb{E}[U] = 0$
- Suppose each loan applicant offered an interest rate R takes the loan offer if R is below a threshold, where the threshold is higher if the applicant's PD is higher
  - Rationale: Riskier applicants are more likely to accept a given offer because their chances to get a loan from another lender are lower
  - Mathematically: each applicant takes the offer if $R < \overline{R} + PD$
  - $\overline{R}$ is the maximum rate a risk-free borrower would accept

Lender's Decisions

- How to predict default?

  1. Construct scoring model: Use data on previous loans' performance to estimate PD as a function of borrowers' observed characteristics: $f(X)$

  2. Score new applications: Best estimate of new application with characteristics X is $\mathbb{P}[D = 1 \mid X] = f(X)$

- How to set the interest rate?

  - Simple idea: take a margin over estimated PD: $R = f(X) + M$ with $M > 0$

Q. What is your expected profit per loan actually granted?

a. M b. more than M c. less than M d. it depends

Statistical Underpinnings of Adverse Selection

- Expected profit per loan granted (conditional on X)

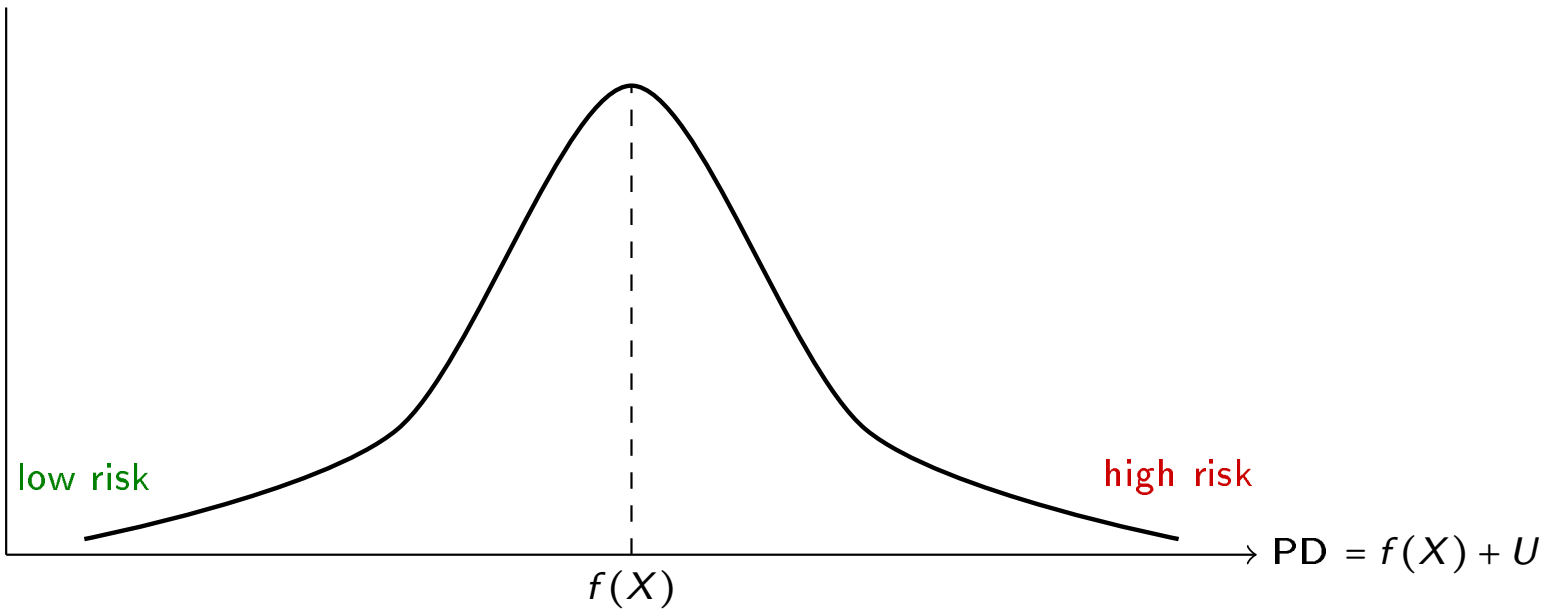
$$
\begin{aligned}
&= \mathbb{E}[R - D \,|\, \text{loan offer is accepted}] \\
&= R - \mathbb{E}[D \,|\, R < \overline{R} + PD] \\
&= f(X) + M - \mathbb{E}[f(X) + U \,|\, f(X) + M < \overline{R} + f(X) + U] \\
&= M - \mathbb{E}[U \,|\, U > M - \overline{R}] \quad < \quad M \quad !!!
\end{aligned}
$$

$\Rightarrow$ You earn less than M per loan. What happened?

- NB: You are not expected to do such calculations at the exam

Statistical Underpinnings of Adverse Selection

Adverse selection! Risky applicants are more likely to accept the loan offer



Statistical Underpinnings of Adverse Selection

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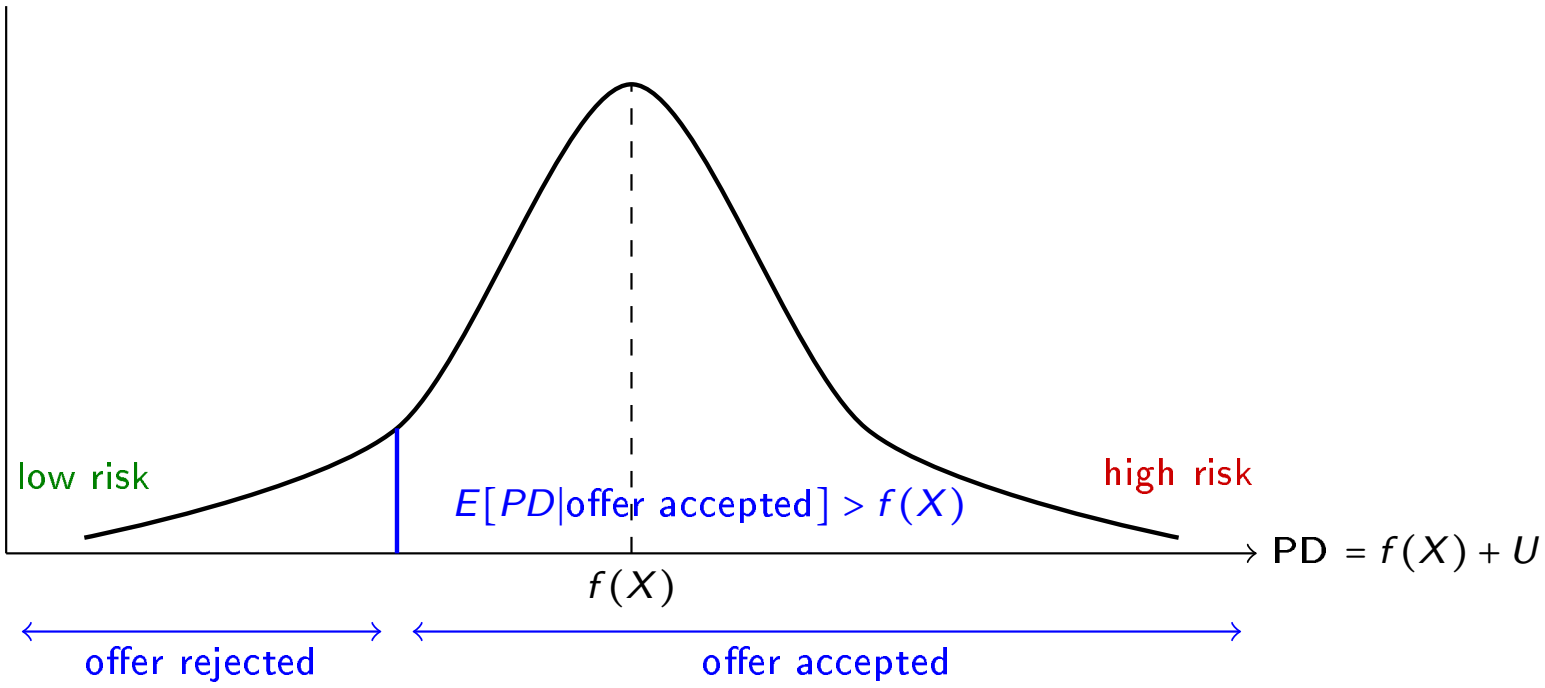
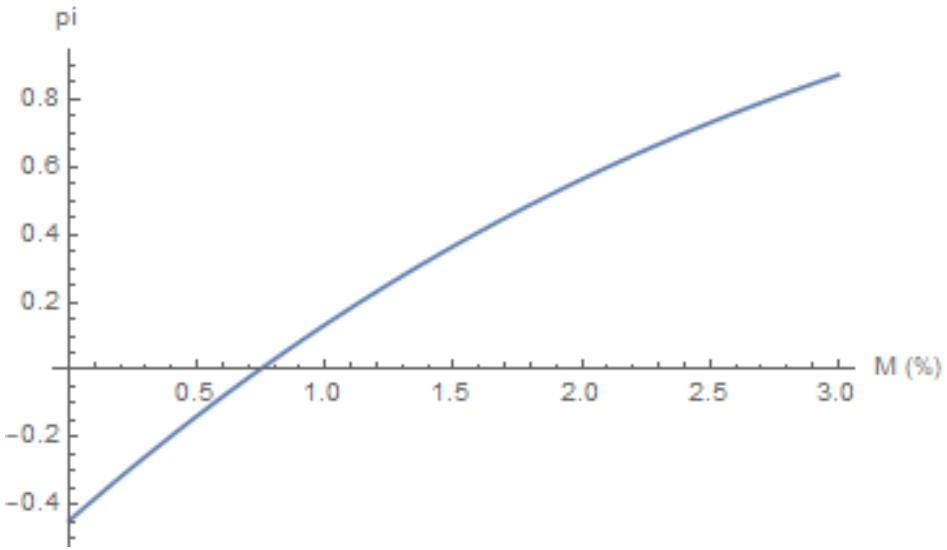


Statistical Underpinnings of Adverse Selection

- Expected profit per loan granted as function of M

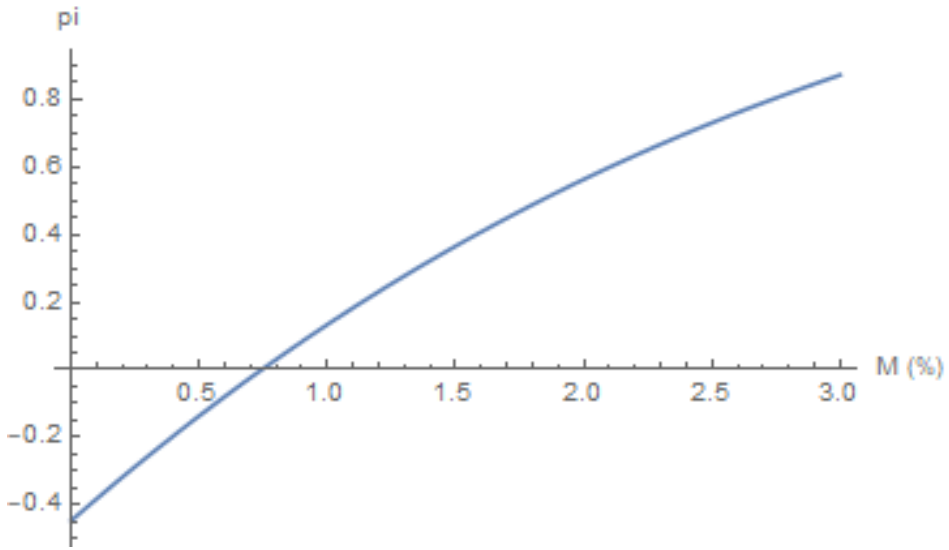


$\overline{R} = 2\%$; $U \sim \mathcal{N}(0, 1.8\%)$

- Adverse selection: profit per loan is less than M and can be negative even if $M > 0$

Statistical Underpinnings of Adverse Selection

- Expected profit per loan granted as function of M



$\overline{R} = 2\%$; $U \sim \mathcal{N}(0, 1.8\%)$

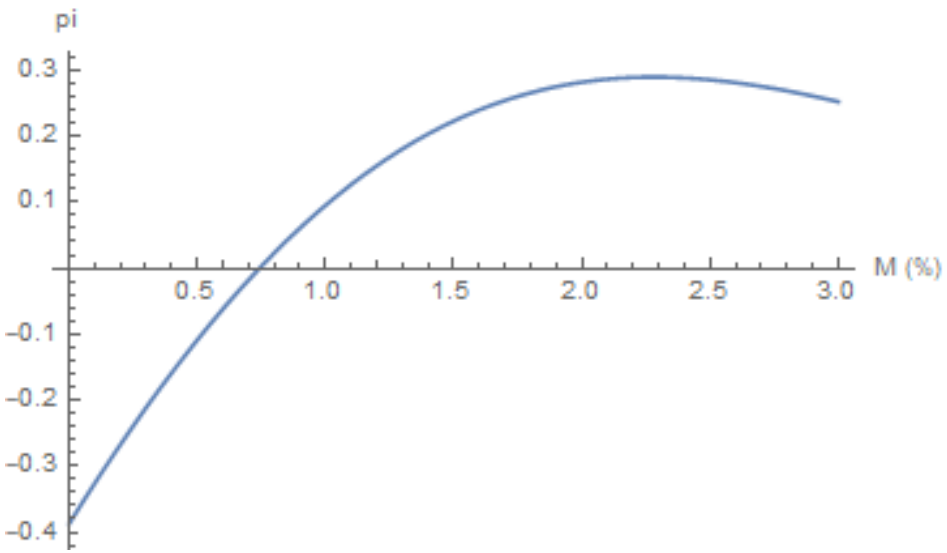
- Adverse selection: profit per loan is less than M and can be negative even if $M > 0$

- Next question: Which M maximizes expected profit?

Statistical Underpinnings of Adverse Selection

- Expected profit

$$= \mathbb{E}[(R-D)\times \mathbb{1}\{\text{offer accepted}\}] = \underbrace{\mathbb{E}[R-D\,|\,\text{offer accepted}]}_{\uparrow M \text{ (previous graph)}} \times \underbrace{\mathbb{E}[\text{offer accepted}]}_{\downarrow M}$$



- Why hump-shaped? Higher $M \Rightarrow$ higher profit per loan granted but fewer loan offers are accepted

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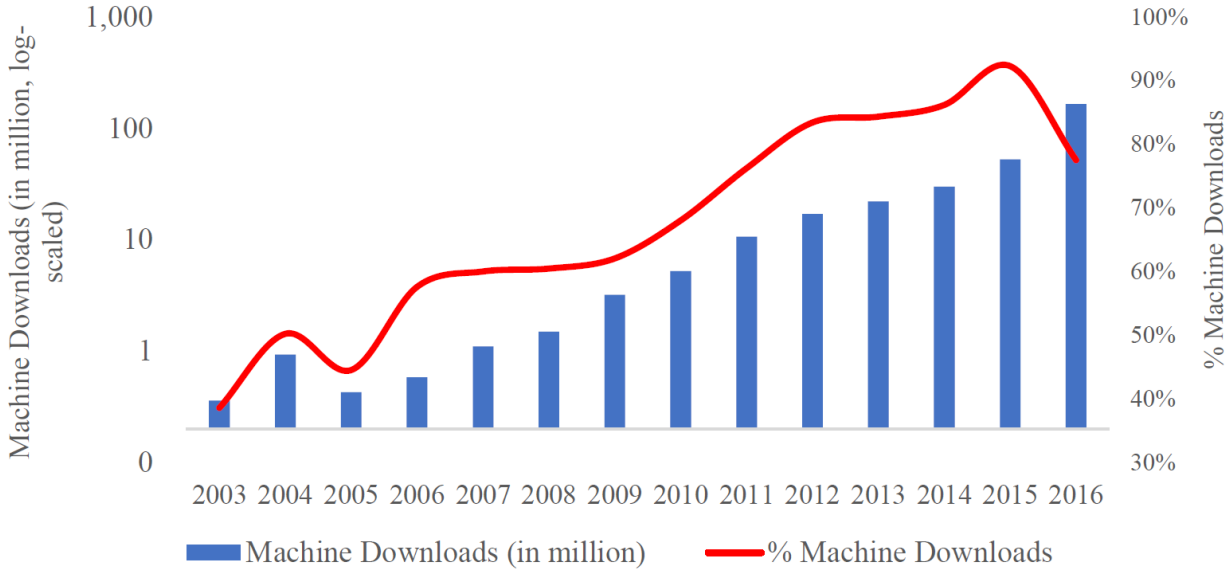
Goodhart's Law

- This figure shows machine downloads of companies' filings with the SEC[2]

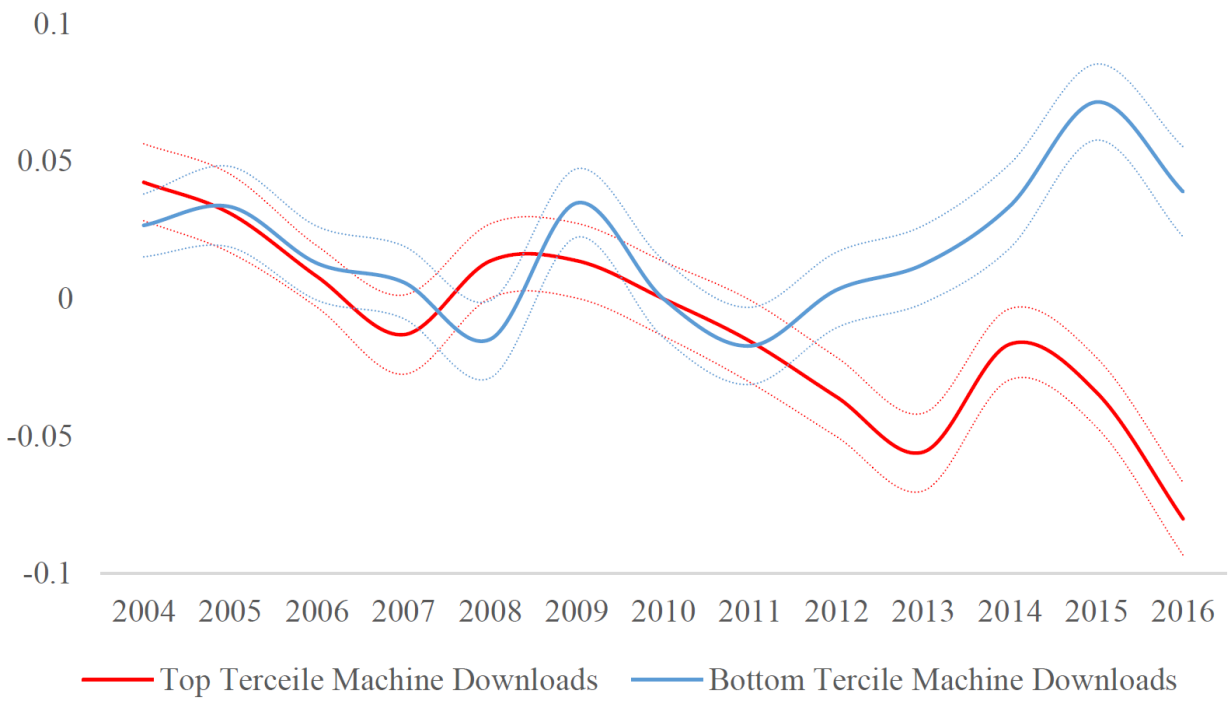


Q. What should companies do?

[2] Cao, Jiang, Yang, Zhang, 2023. "How to Talk When a Machine is Listening: Corporate Disclosure in the Age of AI" *Review of Financial Studies*

Goodhart's Law

- This figure plots the use of words scored negatively by the Loughran-McDonald dictionary (widely used by quant funds before LLMs) by companies analyzed by machines (red) versus companies less analyzed by machines (blue)



$\Rightarrow$ Incentives to use language evaluated positively by algorithms

Q. What is the end game?

Implications of Goodhart's Law

1. Beware scores based on behavior that can be strategically modified

   $\Rightarrow$ Ask yourself if data is exogenous or the result of a strategic choice

2. **Goodhart's law:** A good predictor when unused can become a poor predictor once used

   $\Rightarrow$ Check whether predictive power changes after the data is used

Implications of Goodhart's Law

3. More data can make everyone worse off

   - People being scored try to game the AI ⇒ Costly for them

   - The AI's predictive power decreases ⇒ Costly for companies using AI

4. Commitment <u>not to</u> use data can sometimes create value

   - Example: *"The Value of Privacy: Evidence from Online Borrowers"* Huan Tang (Wharton, HEC PhD 2020)

     Using A/B testing, a lending platform found that customers are more likely to apply for a loan when asked to provide less information

     ⇒ Trade-off between value proposition (privacy) and adverse selection

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Discrimination

- Legal and conceptual distinction between:
  - Direct discrimination
  - Indirect discrimination

Direct Discrimination

- Treatment is based on a protected characteristic such as sex, ethnicity, social origin

- May happen for two reasons:

  1. Outright prejudice

     Example: Job opening for white men only

  2. Statistical discrimination: the protected characteristic is a predictor of risk

     Example: Cheaper car insurance for women because women have fewer accidents

- Illegal in both cases in EU and US

Indirect Discrimination

- Treatment is not based on protected characteristics but ends up being different for people with a protected characteristic

- Happens when treatment is based on variables correlated with protected characteristics

  Example: Interest rate based on borrower's job occupation may end up different for people with different ethnicities

- May be legal or illegal (legal for "business necessity" in US)

AI and Discrimination

- Mortgage loans in the US, 2009–2015[3]
- Lenders sell mortgages to government-sponsored entities ⇒ no retained credit risk
- Minority borrowers pay higher interest rates (controlling for given borrower characteristics)

  +8 basis points per year with traditional lenders

  +5 basis points per year with algorithmic lenders

Q1. What may explain the pattern found with traditional lenders such as human loan officers? Is it direct/indirect discrimination?

Q2. What may explain the pattern found with algorithmic lenders? Is it direct/indirect discrimination?

[3] Bartlett et al., 2021, "Consumer Lending Discrimination in the FinTech Era," *Journal of Financial Economics*

AI and Discrimination

Wells Fargo, Upstart criticized after study finds loan disparities

Published Feb. 6, 2020 • Updated Feb. 14, 2020

The request comes a week after the nonprofit Student Borrower Protection Center found that an Upstart borrower who attended historically black Howard University would pay thousands of dollars more on average for a five-year loan than a borrower with an identical credit profile who studied at New York University.

AI and Discrimination

Fintech Lender: Upstart Network, Inc.
Product: Private Student Loan Refinance

LOAN AMOUNT $30K

Borrower Profile

New York University (Non-MSI)	Howard University (HBCU)	New Mexico State University (HSI)
Major: Computer science	Major: Computer science	Major: Computer science
Occupation: Financial analyst	Occupation: Financial analyst	Occupation: Financial analyst
Annual income: $50,000	Annual income: $50,000	Annual income: $50,000
LOAN OFFERS	LOAN OFFERS	LOAN OFFERS
Loan interest rate: 16.34% APR	Loan interest rate: 21.29% APR	Loan interest rate: 19.23% APR
Origination fee: $1,231	Origination fee: $1,960	Origination fee: $1,862
Total Cost: $42,288	Total Cost: $45,785	Total Cost: $44,011

Q. Is it direct/indirect/not discrimination?

Invisible Primes: Fintech Lending with Alternative Data*

Marco Di Maggio† Dimuthu Ratnadiwakara‡

January 4, 2024

Abstract

We use anonymized data and an underwriting algorithm that incorporates alternative data from a major platform to study how alternative data affects credit access and borrower outcomes. Comparing actual outcomes of the fintech platform's model to counterfactual outcomes based on a "traditional model" used for regulatory reporting purposes, we find that the fintech platform's model approves 15-30% of low credit score applicants rejected by the traditional model and offers substantial reductions in interest rates. The borrowers most positively affected are the "invisible primes"—borrowers with low credit scores and short credit histories, but also a low propensity to default. Some high credit score borrowers are instead rejected by the platform. About two-thirds of the effects are due to the inclusion of additional data, while the remainder is due to a more sophisticated underwriting model. Leveraging exogenous variations in credit access, we show that funding loans to invisible primes leads to better economic outcomes for borrowers.

6th CFPB Research Conference, 5th Annual IMF Macro-Financial Research Conference, Eighth ECB Annual Research Conference 2023, and 4th Future of Financial Information Conference for helpful comments. We thank Upstart Network, Inc. for providing access to its data. Special thanks to Don Carmichael for his assistance in setting up access to the data infrastructure on the cloud and for his assistance in navigating

Q. How can indirect discrimination against racial minorities be consistent with benefiting "invisible primes"?

Discrimination and Redistributive Effects

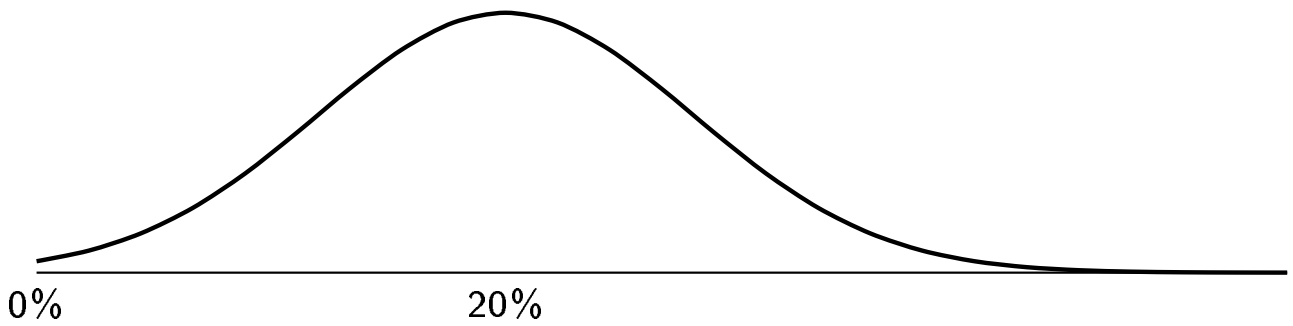
- Consider the same theoretical framework as previously
  - You make loans with interest rate R, paid at issuance
  - CF at issuance: $-1 + R$ (per euro lent)
  - CF at maturity: 1 if the borrower repays
    0 if the borrower defaults
- Borrowers have a 20% average default rate
- They are willing to pay a 8% interest rate maximum

Q1. Can you profitably make loans without additional information about loan applicants? a. yes b. no

Discrimination and Redistributive Effects

- Now, you have access to alternative data about loan applicants ⇒ Credit scoring model gives you, for each loan applicant k, an estimated probability of default PD_k

- The distribution of estimated PD has mean 20% and the following shape:

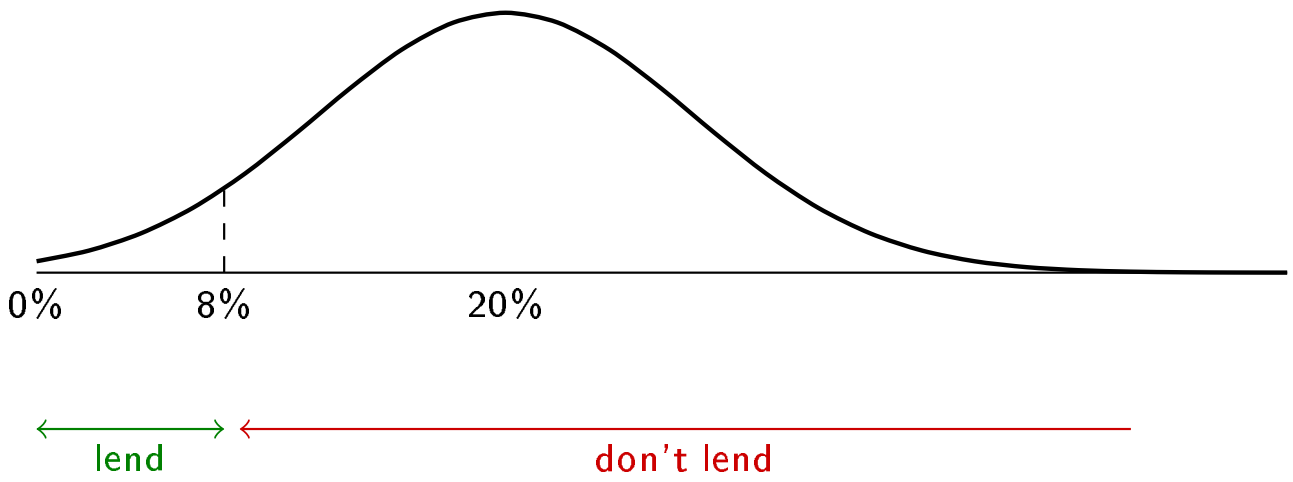


Q2. Can you now profitably make loans? a. yes b. no

Discrimination and Redistributive Effects

- Now, you have access to alternative data about loan applicants ⇒ Credit scoring model gives you, for each loan applicant k, an estimated probability of default PD_k

- The distribution of estimated PD has mean 20% and the following shape:



Q2. Can you now profitably make loans? a. yes b. no

Q3. Is the use of the additional data beneficial to everyone? a. yes b. no

Discrimination and Redistributive Effects

1. "Invisible primes" gain financing: loan applicants initially pooled in the 20% default risk pool, whose data reveal positive information

2. Business opportunity: lender can charge 8% to these loan applicants

$\Rightarrow$ Beneficial for everyone

Discrimination and Redistributive Effects

- Consider a different population of loan applicants with lower average risk
- Without data: 5% average default probability
- Loan applicants are still willing to pay up to 8%

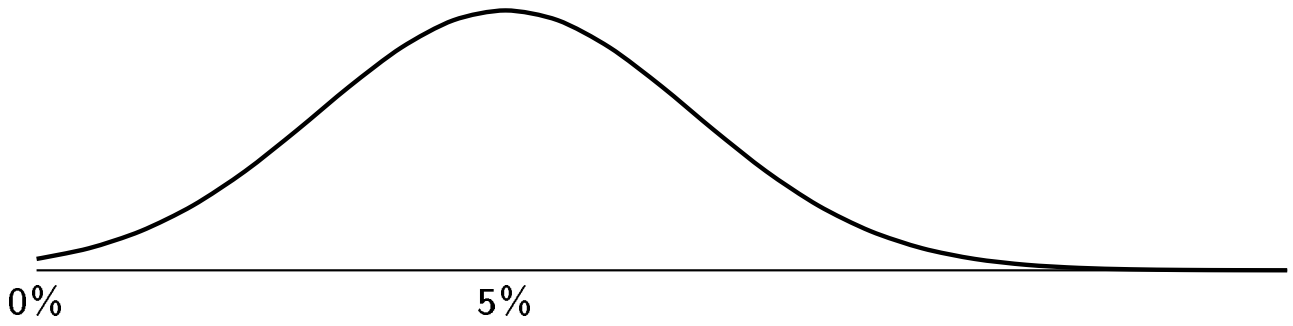
Q4. Can you profitably make loans without additional information about loan applicants? a. yes b. no

Discrimination and Redistributive Effects

- Consider a different population of loan applicants with lower average risk
- Without data: 5% average default probability
- Loan applicants are still willing to pay up to 8%

Q4. Can you profitably make loans without additional information about loan applicants? a. yes b. no

- With additional data: risk estimate PD_k specific to each loan applicant



Q5. How do you change your lending policy relative to Q4? Is the use of the additional data beneficial to everyone? a. yes b. no

Discrimination and Redistributive Effects

- Consider a different population of loan applicants with lower average risk
- Without data: 5% average default probability
- Loan applicants are still willing to pay up to 8%

Q4. Can you profitably make loans without additional information about loan applicants? a. yes b. no

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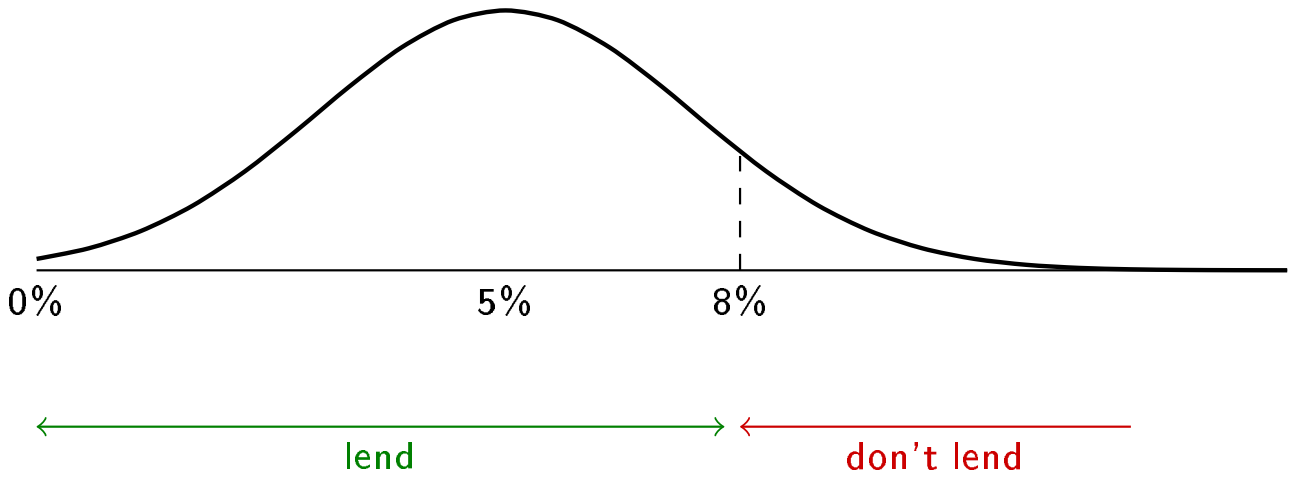

Q5. How do you change your lending policy relative to Q4? Is the use of the additional data beneficial to everyone? a. yes b. no

Discrimination and Redistributive Effects

3. Some people lose financing: those initially pooled in the 5% default risk pool, whose data reveal negative information

4. Additional data improves the allocation of capital in the economy: more credit to people with low default probability

$\Rightarrow$ **Tradeoff between allocative efficiency and redistributive issues**

- See practice problem 1

Road Map

Mini-Case 1: Credit Scoring

VC Game

Mini-Case 2: Zillow Offers

Statistical Underpinnings

Goodhart's Law

Discrimination and Redistributive Effects

Hirshleifer Effect

Information in Insurance

- Suppose we discover how to predict perfectly who will get sick, but this foreknowledge does not help to prevent or cure diseases

- Insurers use this information to price health insurance

Q1. Will this make people better or less-well insured?

a. better insured b. less-well insured

Q2. Will this make insurers more or less profitable?

a. more profitable b. less profitable

Information in Insurance

- What will happen?
  - No insurers accept to insure people predicted to be sick
  - People predicted to be healthy don't need insurance

$\Rightarrow$ The health insurance market breaks down

  - People are worse off: they can't get insurance
  - Insurers are worse off: they can't sell insurance

- **Hirshleifer effect:** Information can destroy insurance

How to Overcome the Hirshleifer Effect?

- Suppose an insurer announces it will not use the information and the promise is credible

Q3. Does this overcome the Hirshleifer effect and allow the insurer to sell insurance?

a. yes b. no

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Q3. Does this overcome the Hirshleifer effect and allow the insurer to sell insurance?

a. yes b. no

- No, because of adverse selection
  - People predicted to be healthy are offered cheap insurance from other insurers, or they don't even buy insurance
  - The insurer only gets people who will be sick, so it cannot insure them
- See practice problem 2

How To Overcome the Hirshleifer Effect?

- Solution 1: Ensure no insurer uses the information
  - Industry self-regulation
  - Regulation (insurers are not allowed to use DNA testing)
- Solution 2: Insure before information is revealed
  - Long-term insurance
  - Government-provided insurance

Group Work: Lending Game

- 3 lenders (teams of students) compete to make loans
- Data on past loans: borrower information and default
- New loan applications
- All teams have data on the same past loans and receive the same new loan applications, but each team has <u>different</u> borrower information
- The game (detailed guidelines on BB)

Now: Form teams of 4-5 persons and email your team's composition at hombert@hec.fr

Round 1: Each team makes an offer to every loan applicant → offers due on April 18

Each loan applicant chooses which team's loan offer to take. Borrowers repay or default → profits and losses for each team

Round 2: Each team improves its strategy based on performance in round 1 and play the same game again → offers + report due on May 9

- Email me your team composition

- First round of the game due on April 18

- Problem set on today's class on Blackboard

Have a great week ☺